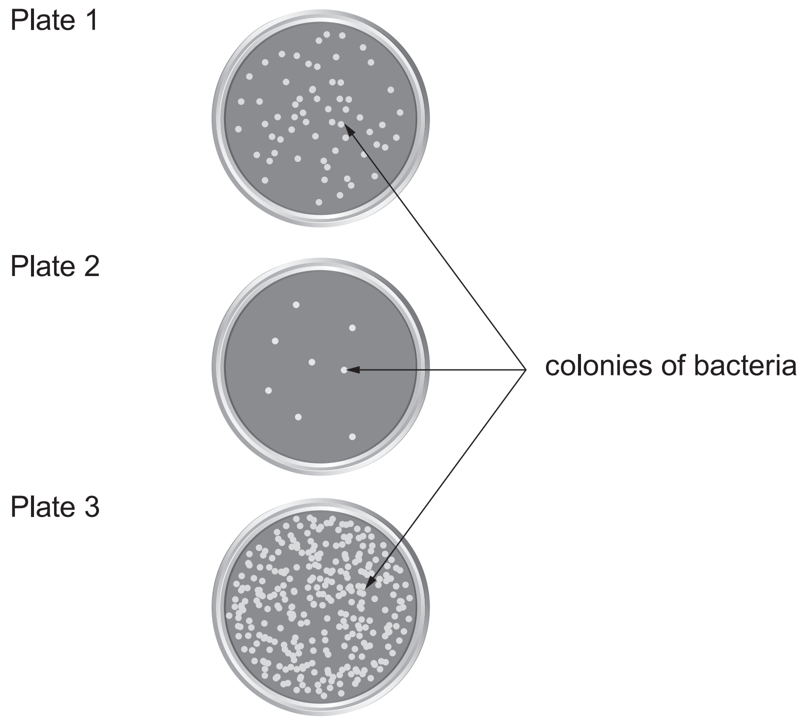
Examiner only

4. A laboratory technician tests **three** different food samples taken from a local restaurant by an Environmental Health Officer.

The samples were spread onto sterile agar plates which were then incubated at 37 °C.

After 3 days the agar plates were examined. The results are shown below.

It is assumed that each colony was formed from one bacterium.



(a) (i) The technician concludes that the food that produced Plate 1 was stored in the restaurant at 15 °C, the food that produced Plate 2 was stored at 25 °C, and the food that produced Plate 3 was stored at 5 °C. Explain whether you agree. [3]

....

....

....

....

....

(ii) State why it is not possible to make a conclusion on the reproducibility of this analysis. [1]

....

....

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(b) Agar plates can also be used in a school laboratory to investigate bacterial growth.

(i) State why the agar plate needs to be sterile. [1]

....

(ii) State what should be done to the inoculating loop before it is used to transfer the food sample to the agar plate. [1]

....

(iii) Give a reason why the students are only allowed to incubate their plates at 25 °C. [1]

....

(iv) State why it is important the lid remains on the agar plate during incubation. [1]

....

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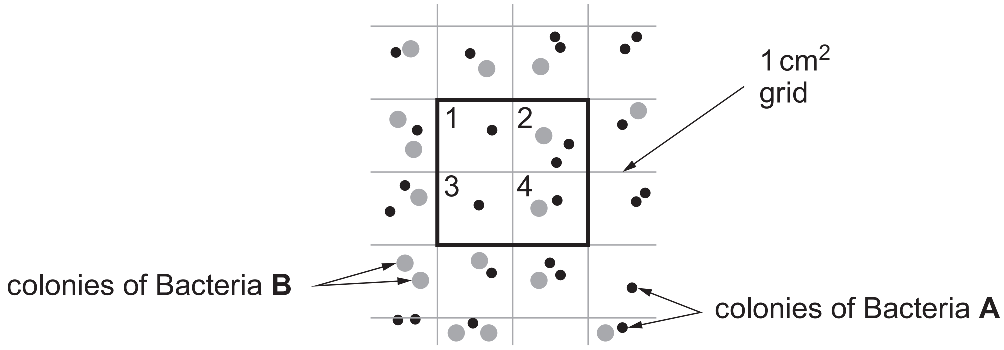
(c) Two types of bacteria, **A** and **B**, were found on one of the plates. The plate has an area of 52 cm^2. Each square in the grid is 1 cm^2.

Section of agar plate



The results for four squares numbered **1** to **4** are shown in the table below. The students have completed the table for Bacteria **A**.

Grid number	Number of bacteria in 1 cm^2	
	Bacteria **A**	Bacteria **B**
1	1	0
2	2	1
3	1	0
4	1	....
Mean colonies per cm^2	1.25	....
Mean colonies per **52 cm^2** plate	65	26
Total number of colonies in squares 1–4	5	....

(i) Complete the table for Bacteria **B**. [3]

(ii) State how the results would be improved by counting the number of bacteria in 12 squares. [1]

....

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